



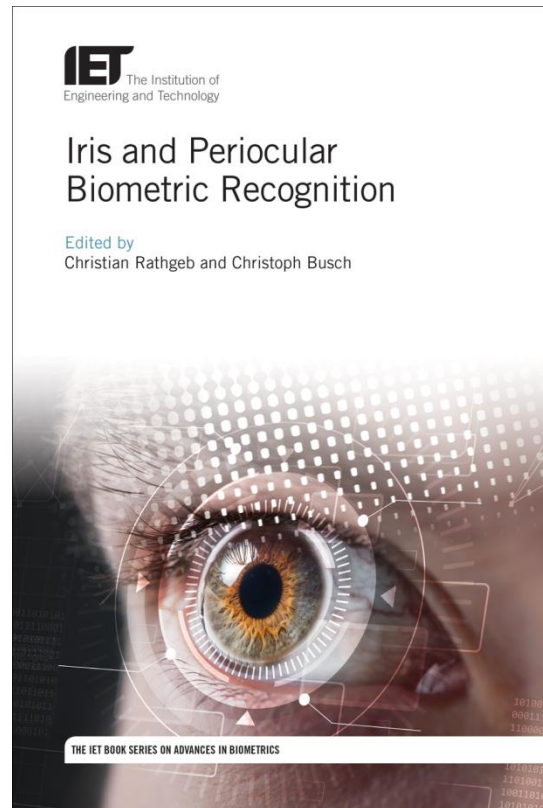
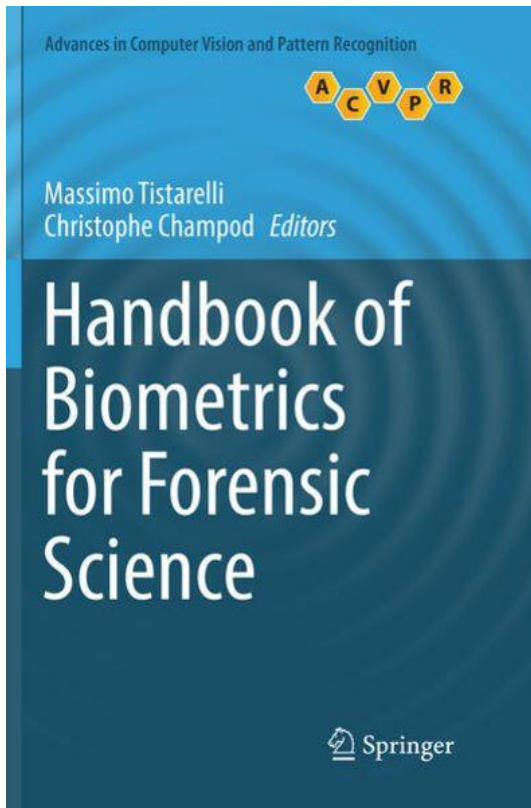
Biometric forensics

Taras Holotyak

Content

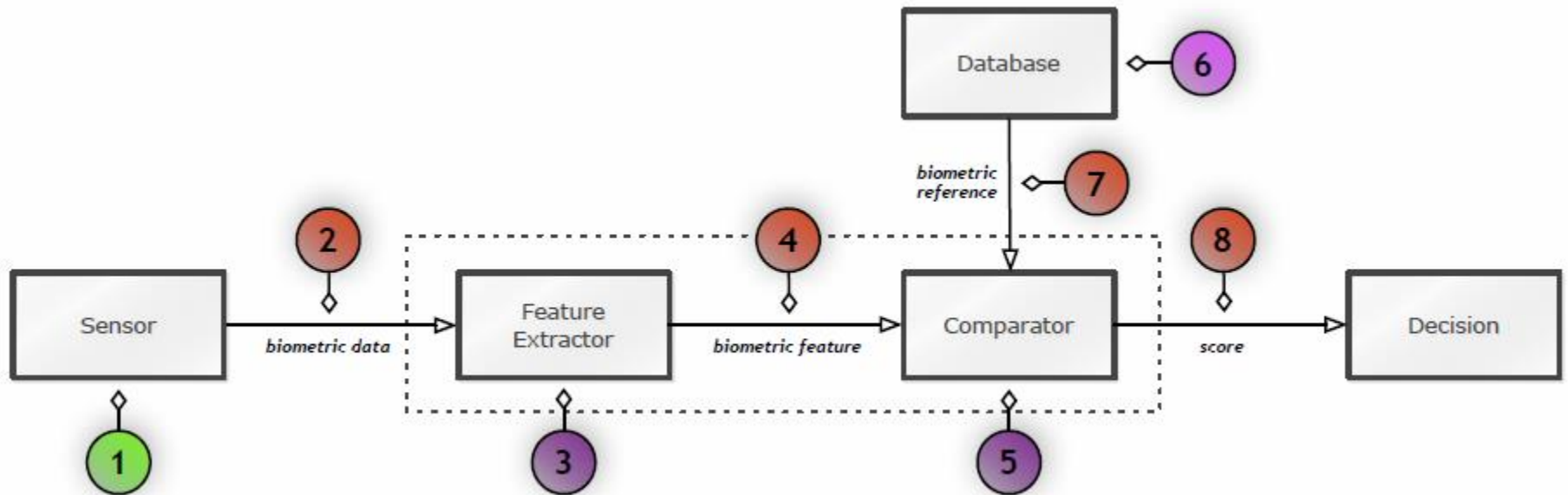
- Introduction
- Presentation attacks
- Defense from face presentation attacks

Introduction



Introduction

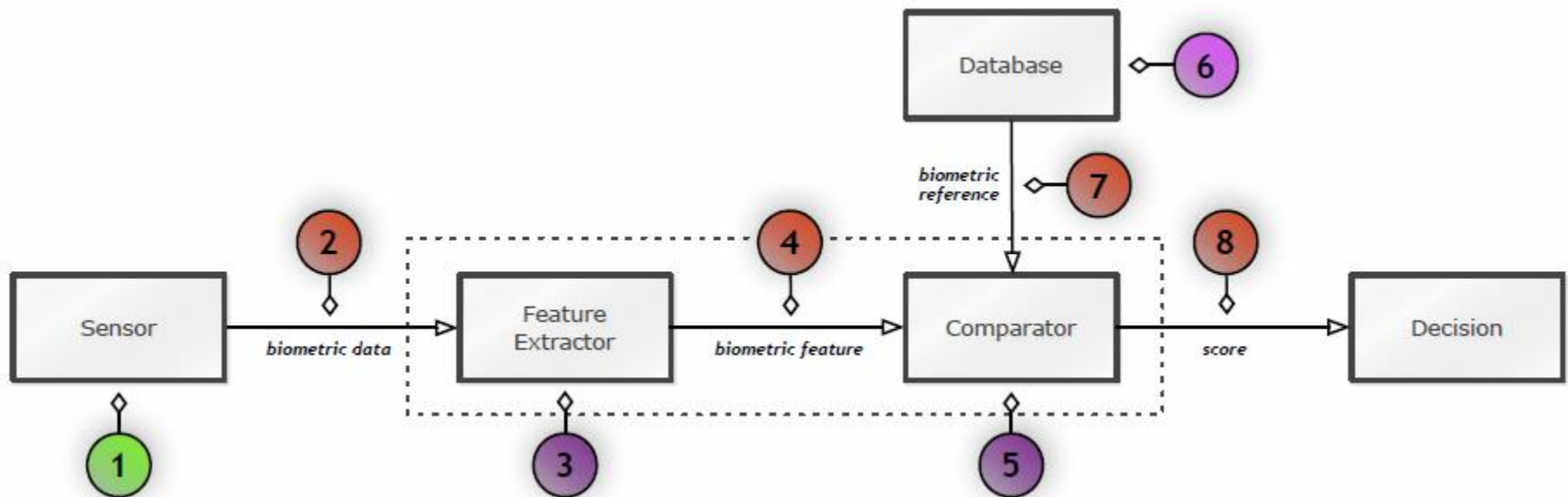
Attacks on biometric systems



- Direct attacks (1)
- Indirect attacks (2-8)

Introduction

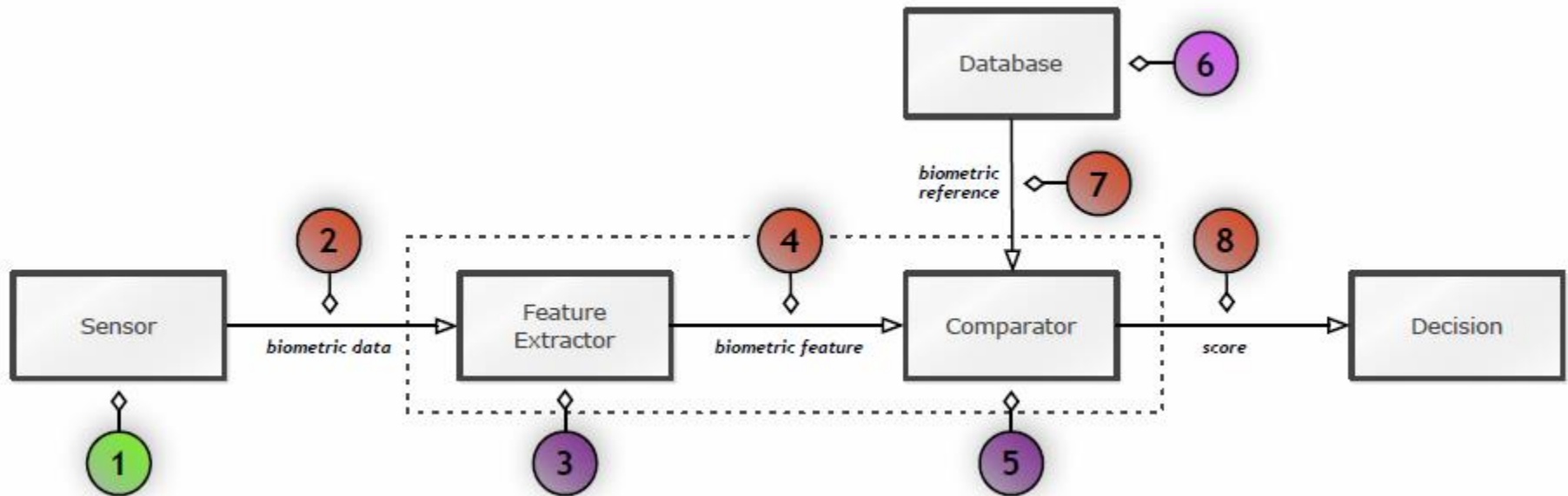
Attacks on biometric systems: direct attacks



Direct attacks (spoofing or presentation attacks – PAs) are performed at the sensor level: the sensor is fooled but not replaced or tampered.

Introduction

Attacks on biometric systems: indirect attacks



Indirect attacks are performed inside the system by:

- bypassing the feature extractor or the comparator (3, 5)
- manipulating the biometric references in the biometric reference database (6)
- exploiting possible weak points in communication channels (2, 4, 7, 8)

Introduction

Examples of presentation attacks



Usage of silicon mask

Introduction

Examples of presentation attacks



Usage of silicon mask

Introduction

Examples of presentation attacks



Usage of silicon mask

Introduction

Examples of presentation attacks



Usage of silicon mask

Introduction

Examples of presentation attacks



Photos as FaceID

Introduction

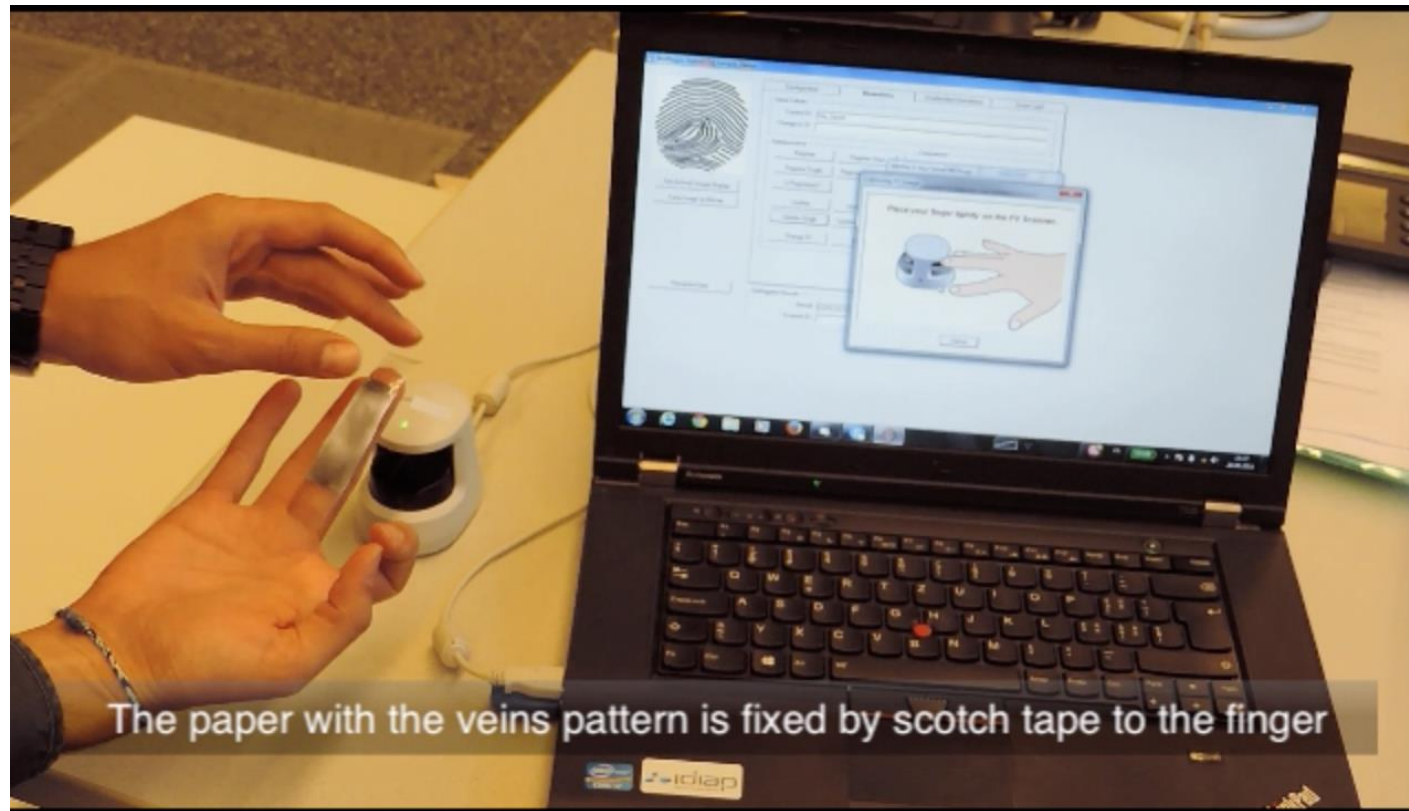
Examples of presentation attacks



Usage of fake fingers

Introduction

Examples of presentation attacks



The paper with the veins pattern is fixed by scotch tape to the finger

Finger-vein spoofing

Introduction

Examples of presentation attacks



Iris spoofing

Presentation attacks

“Gummy Fingers”

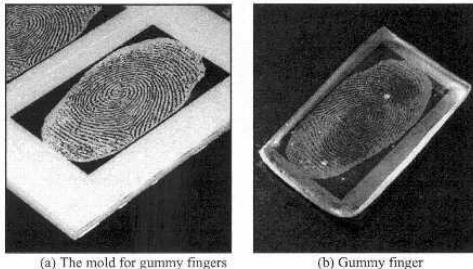


Figure 4.5 Photographs of the outside appearance of the mold and a *gummy* finger. The *gummy* finger was produced from a residual fingerprint on a glass plate, enhancing it with a cyanoacrylate adhesive



Figure 4.6 The Fingerprint image of the *gummy* finger, which was displayed by the system with Device H (equipped with a capacitive sensor).

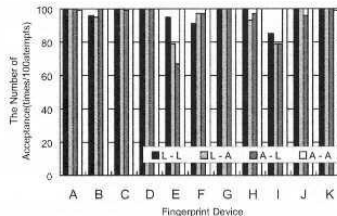
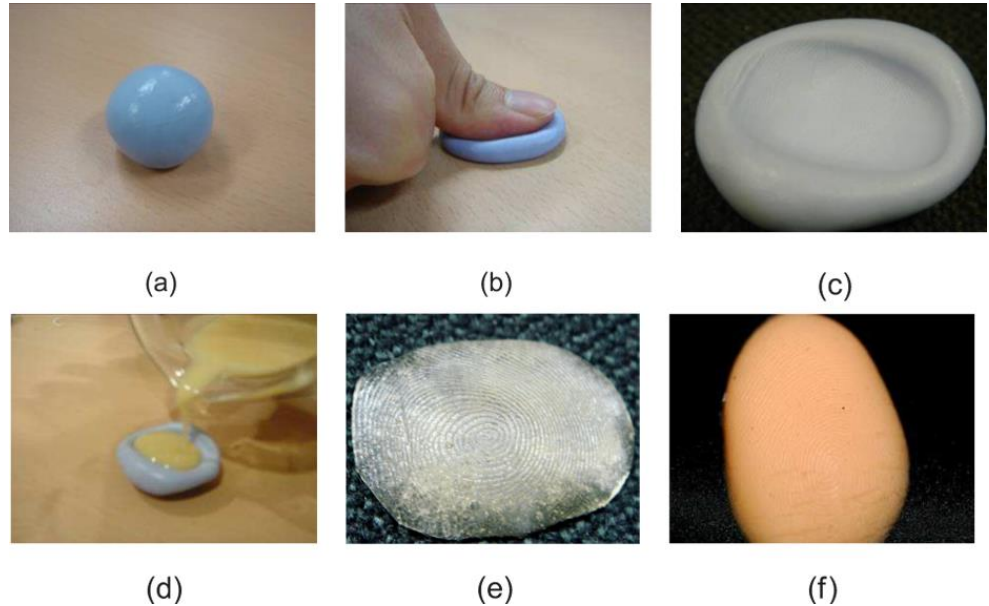


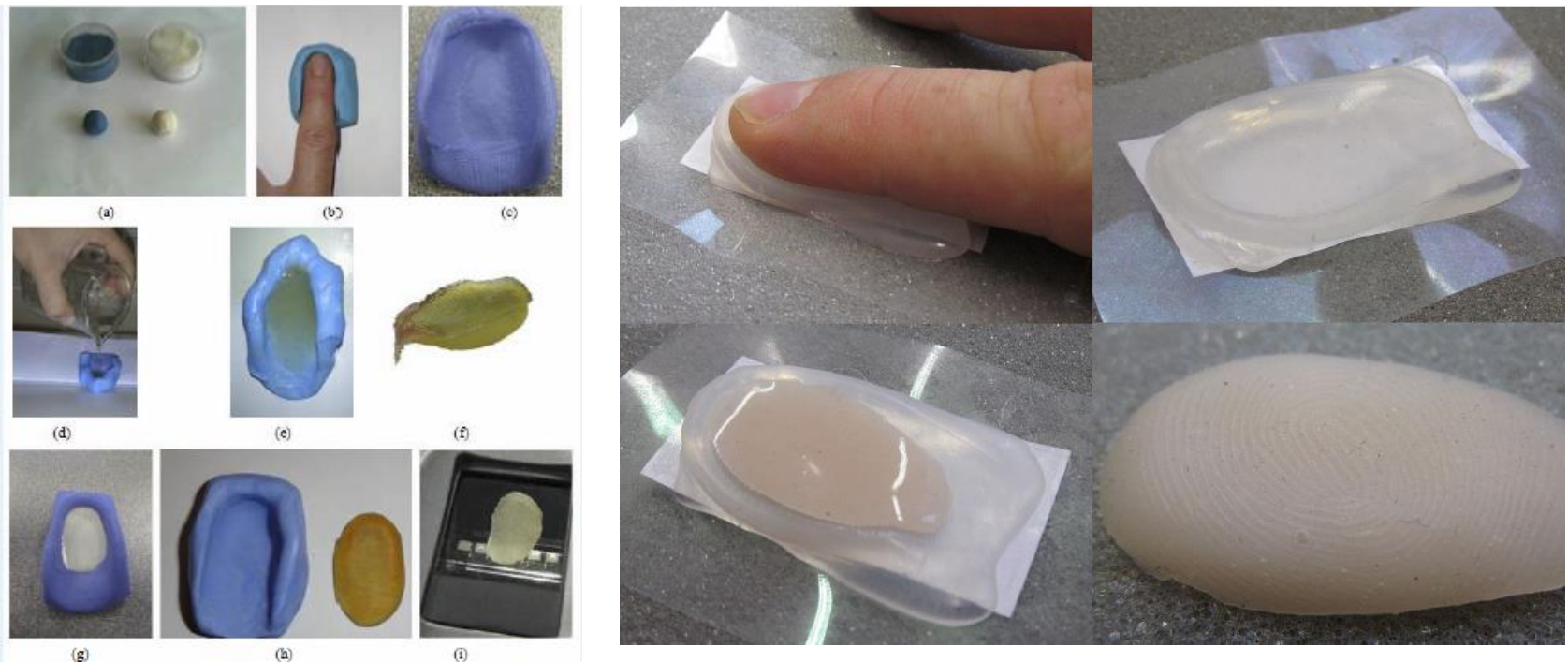
Figure 4.7 Average number of acceptance for each device, in terms of *gummy* fingers which were cloned from residual fingerprints. Here, the subject is one person.



Gelatin fake fingers are used to spoof fingerprint biometric systems

Presentation attacks

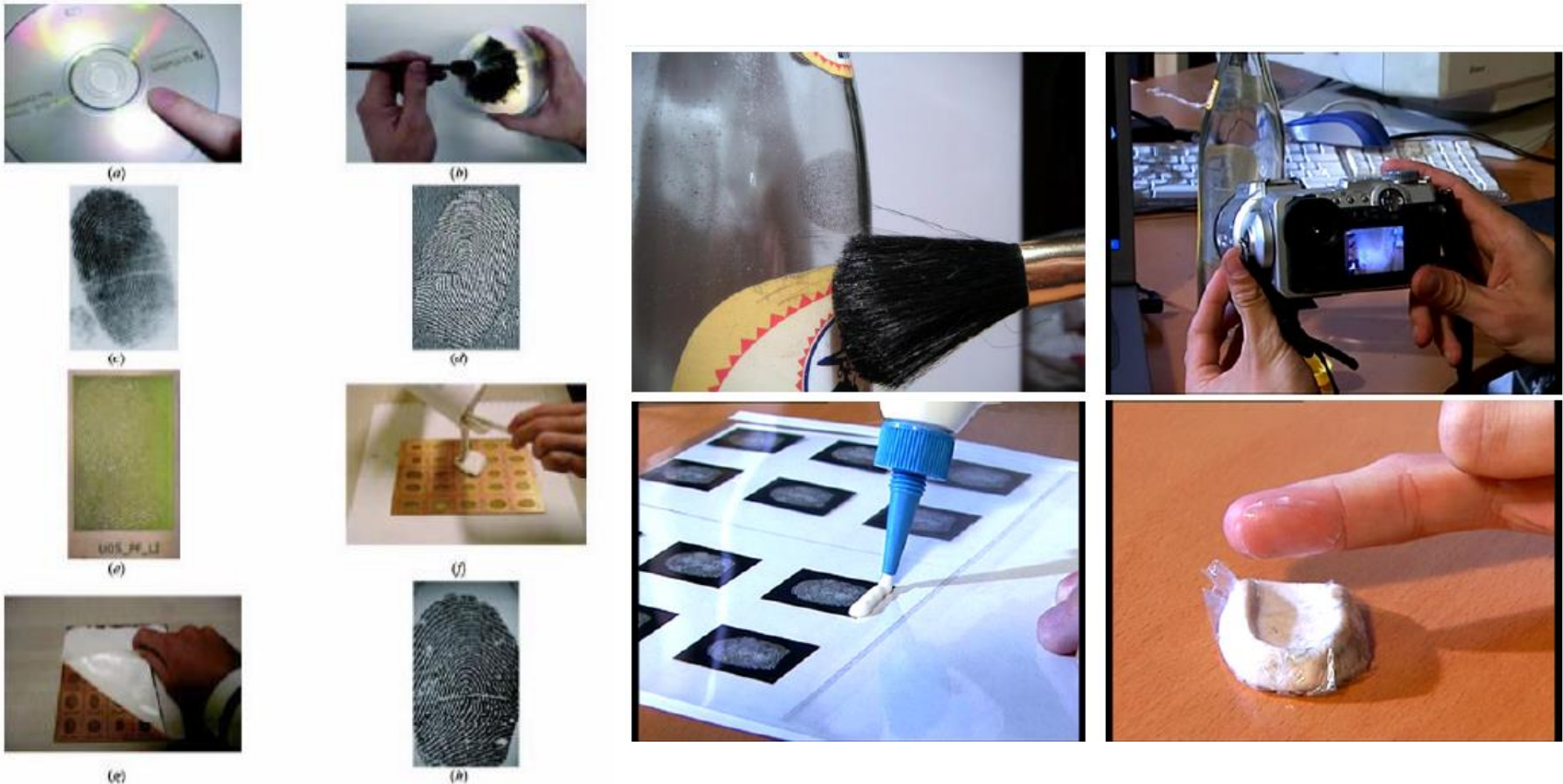
Fingerprint presentation attacks (with cooperation)



Gelatin or wood glue are used to produce fake fingers for spoofing of fingerprint biometric systems

Presentation attacks

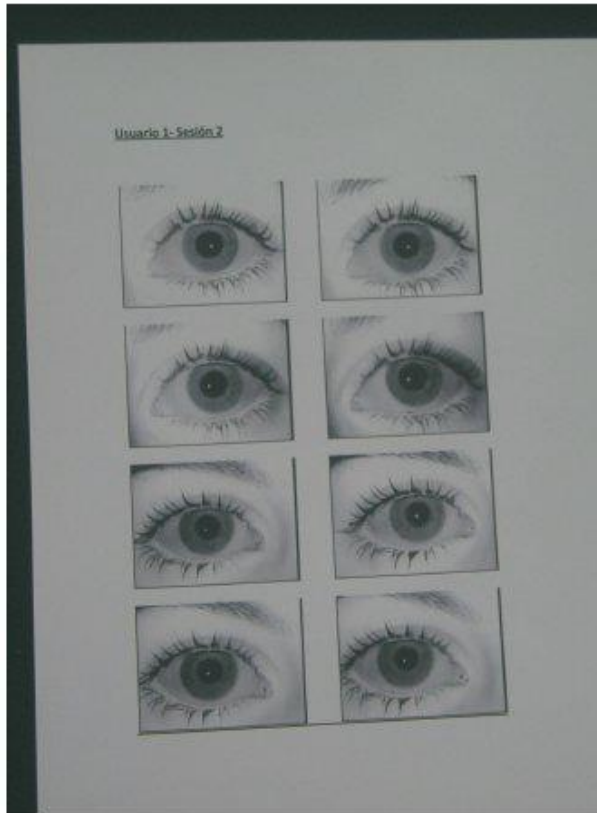
Fingerprint presentation attacks (without cooperation)



Lifted latent fingerprint is printed on a PCB (Printed Circuit Board) to serve as a mold

Presentation attacks

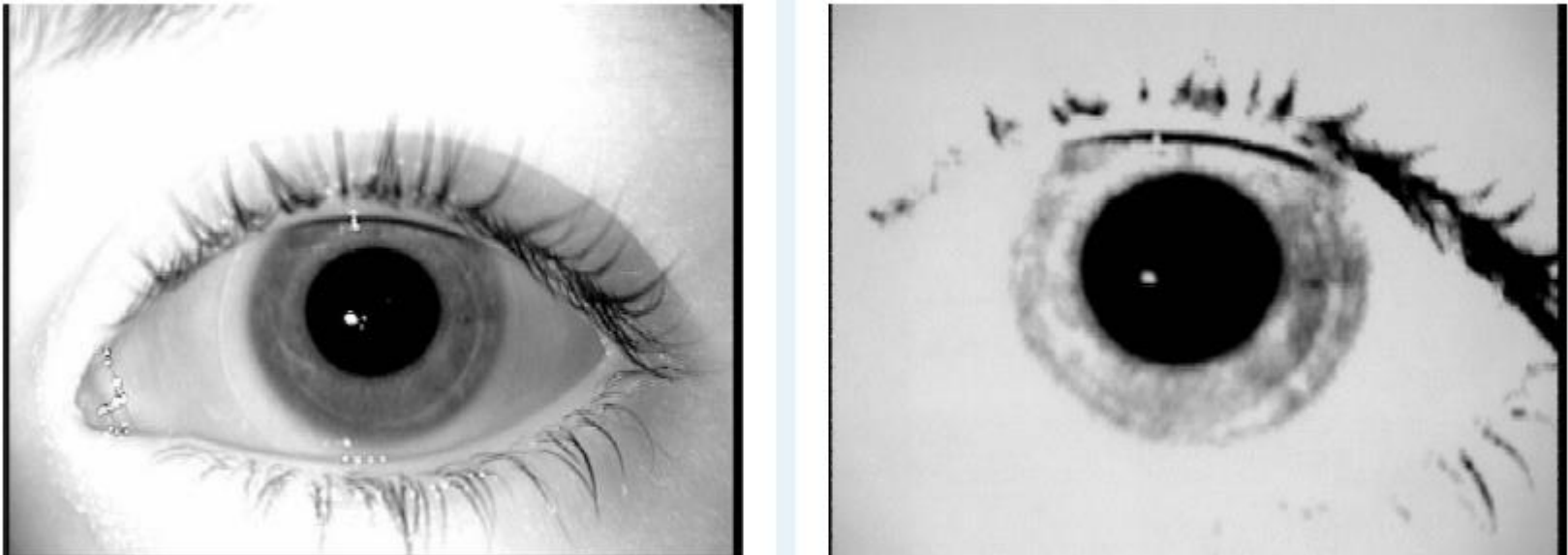
Iris presentation attacks



High quality paper and inkjet printer

Presentation attacks

Iris presentation attacks



Real (left) vs fake (right) iris

Presentation attacks

2D face presentation attacks

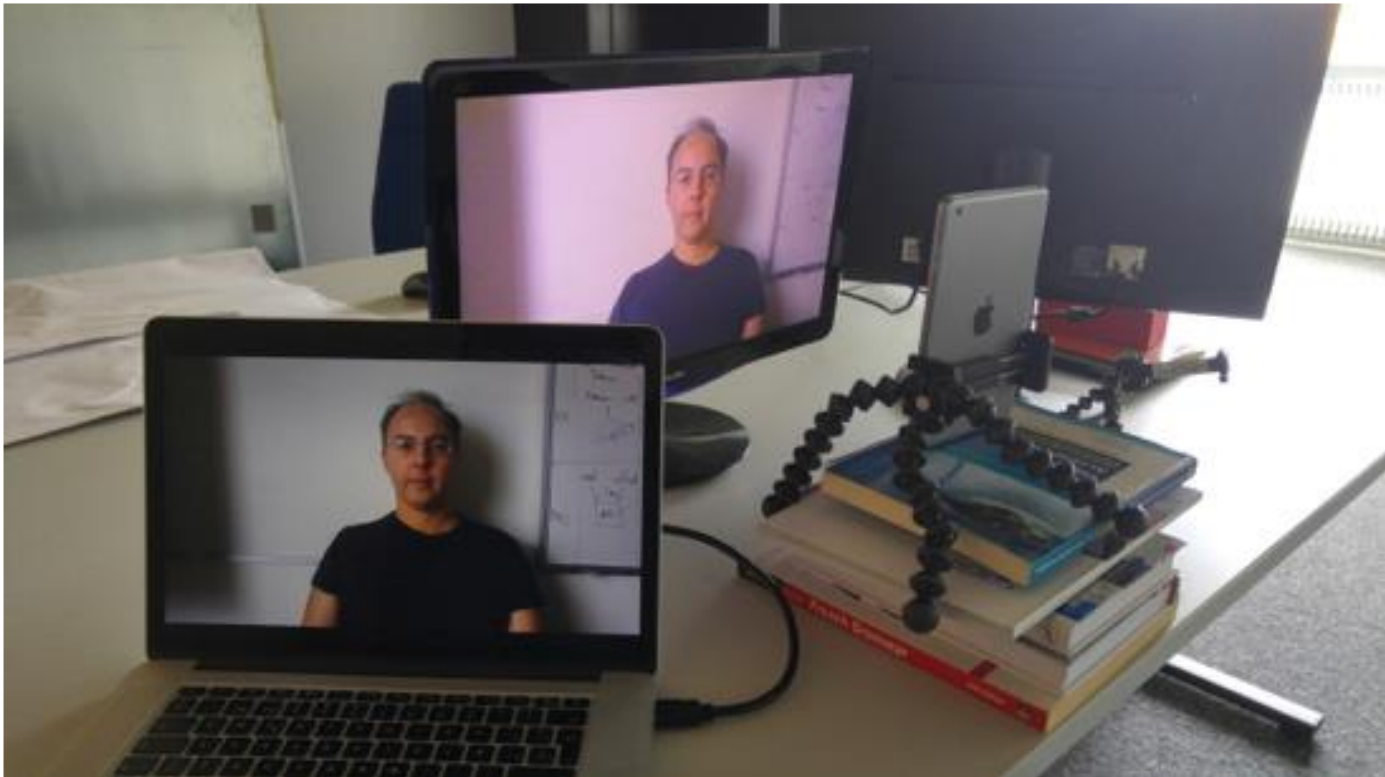


Printed photo to spoof face recognition systems on 3 laptops:

- Asus (F6S Series, X80 Series): Asus SmartLogin ver 1.0.0005
- Toshiba (L310, M300): Toshiba Face Recognition ver 2.0.2.32
- Lenovo (Y410, Y430): Lenovo Veriface III

Presentation attacks

2D face presentation attacks



Presentation attacks

2D face presentation attacks



3D hard resin/silicon mask as attacking tool

Presentation attacks

2D face presentation attacks



3D hard resin/silicon mask as attacking tool

Presentation attacks

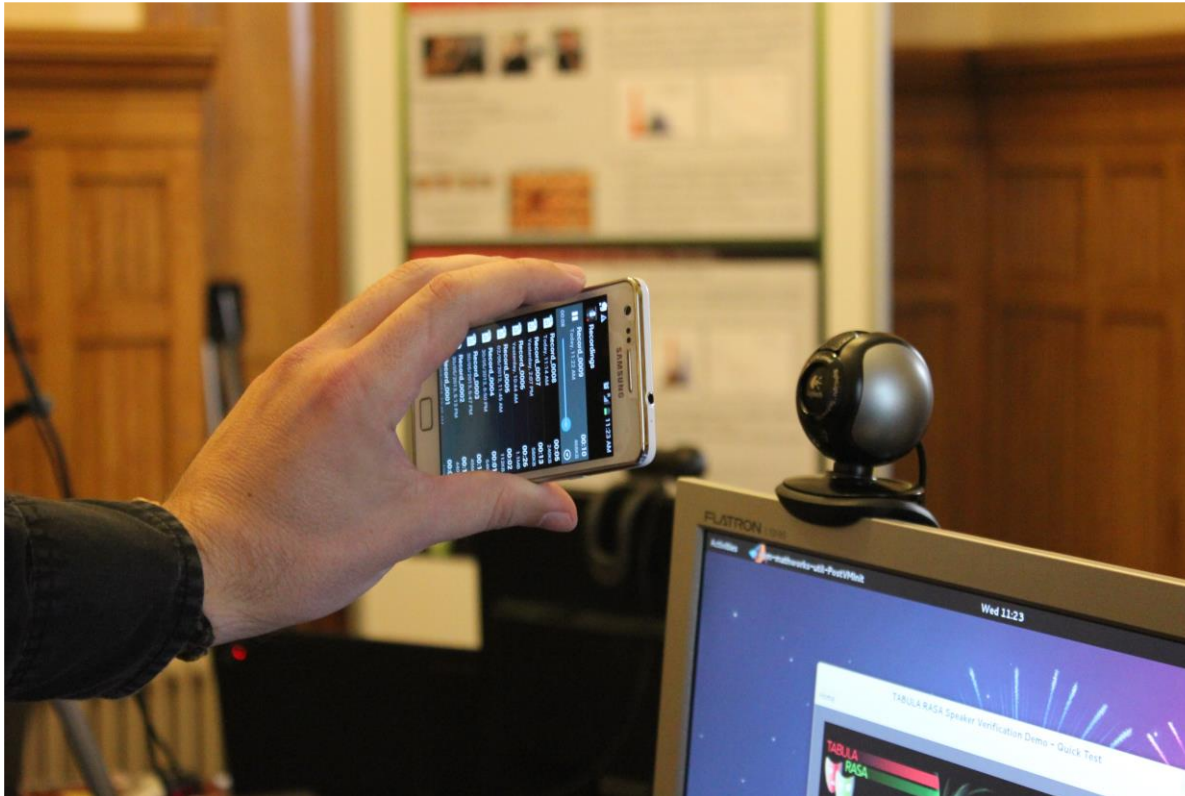
Deepfake 2D face presentation attacks



Realistic face images are generated using a pre-trained Generative Adversarial Network (GAN)

Presentation attacks

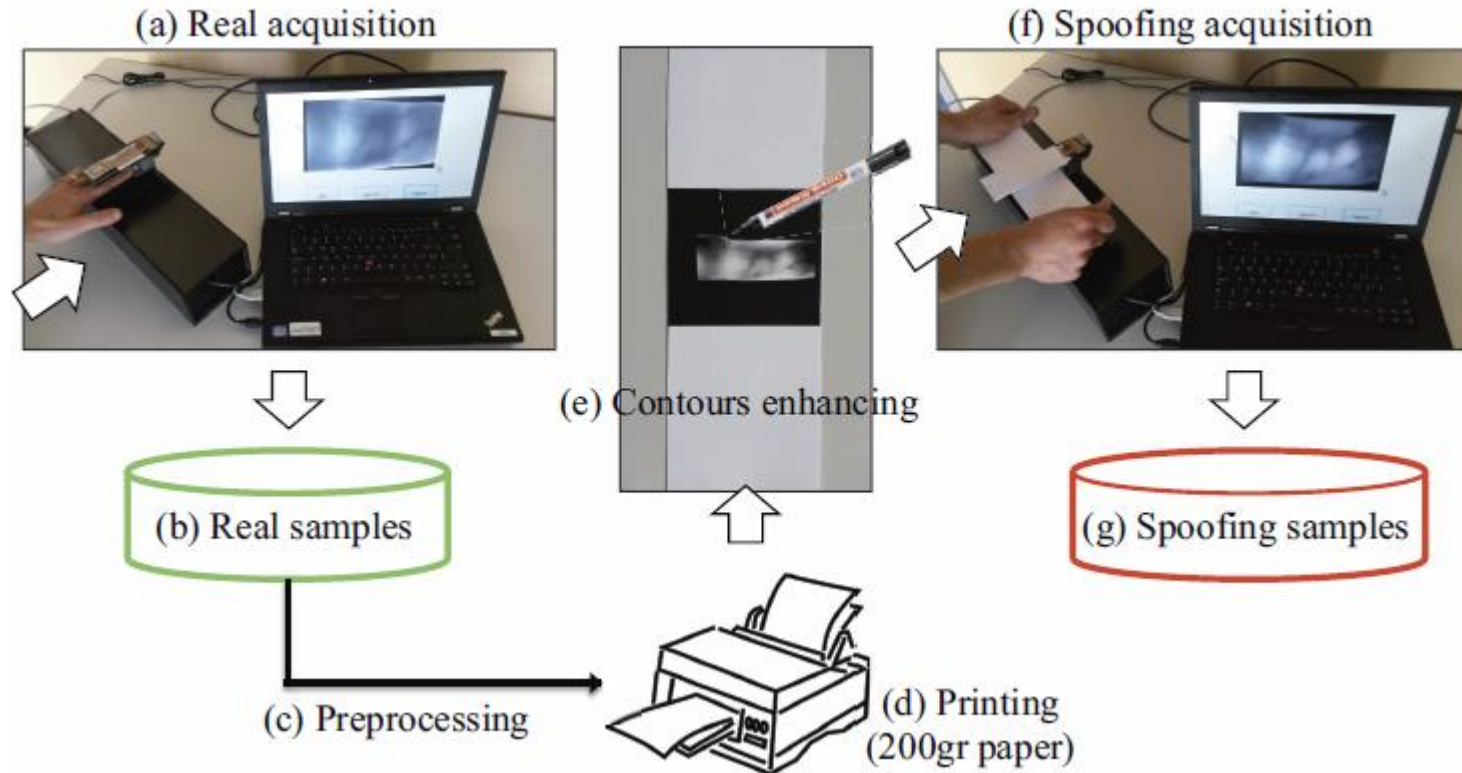
Voice presentation attacks



Playback of a voice recording, a synthesised speech or a converted voice in front of a microphone

Presentation attacks

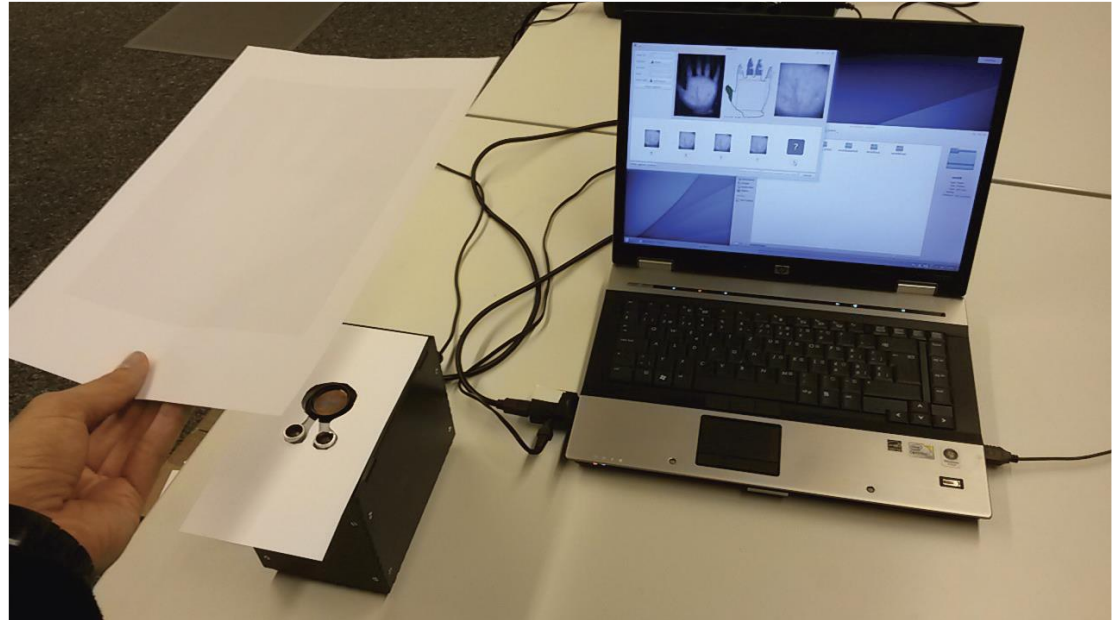
Finger-vein presentation attacks



Playback of a voice recording, a synthesized speech or a converted voice in front of a microphone

Presentation attacks

Palmvein presentation attacks



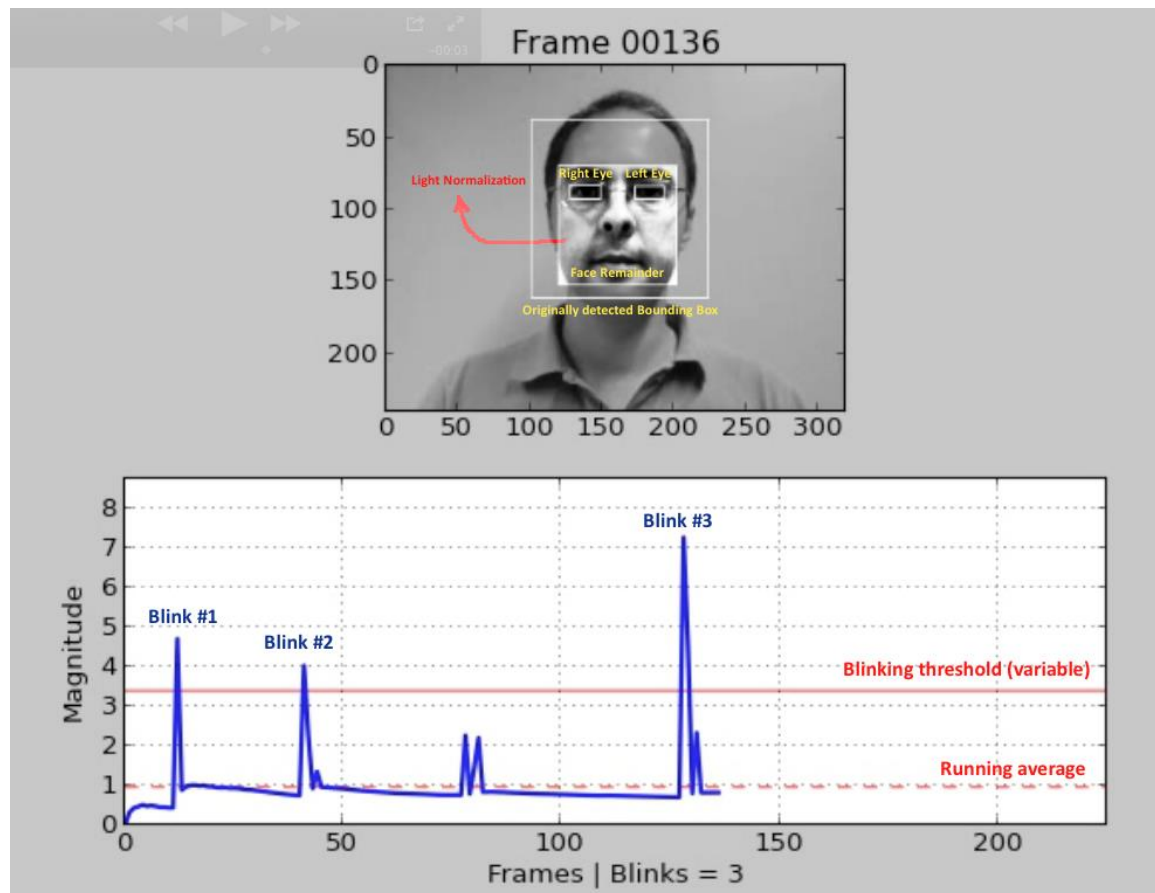
Defense from face presentation attacks

Presentation attack defense methods:

- software-based: biometric data from the sensor is analyzed to discriminate original/alive vs attacked sample (e.g., motion, texture)
- hardware-based: an additional sensor is used and its data analyzed to discriminate original/alive vs attacked sample (e.g., temperature, pulse)
- challenge-response: the user interacts with the system (e.g., prompted text in face/speaker recognition)

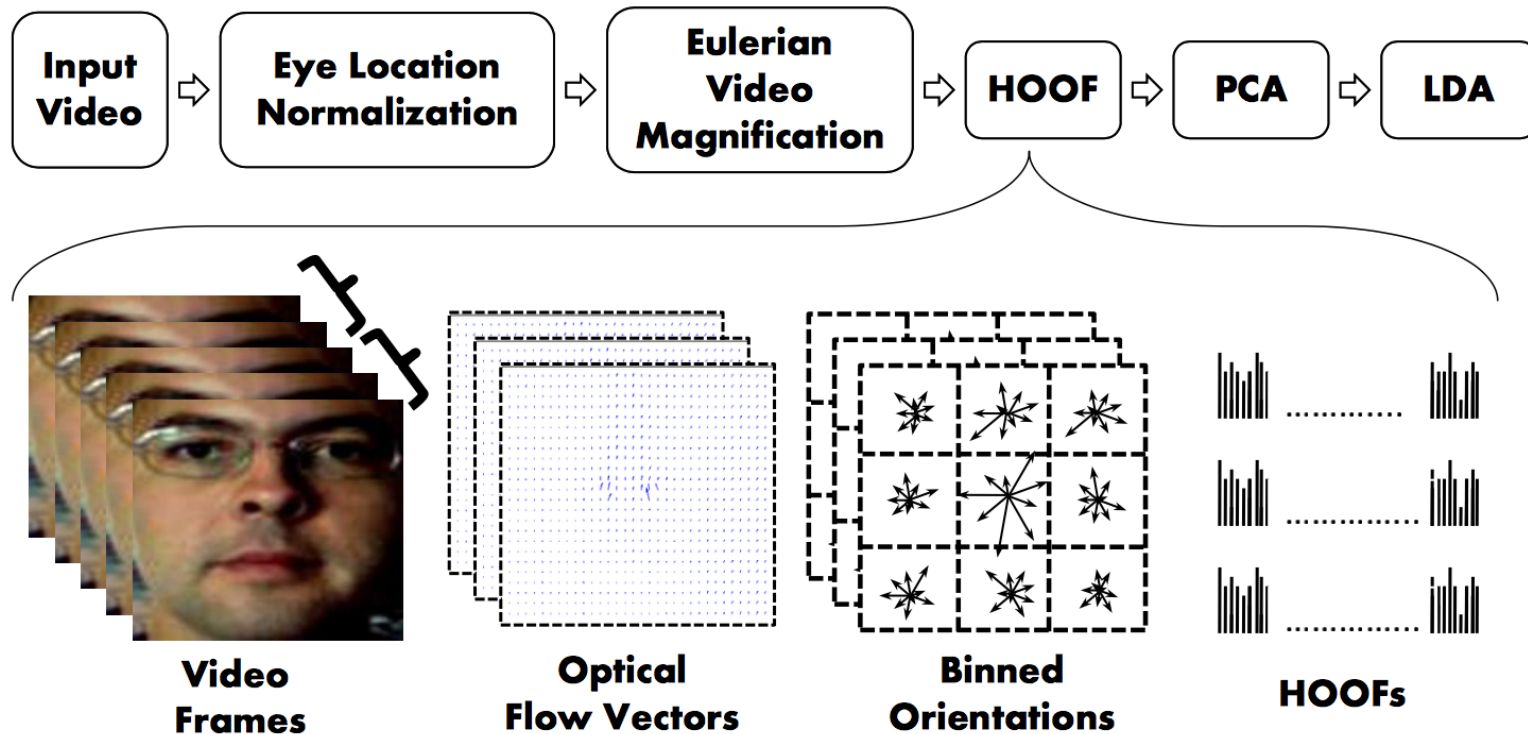
Defense from face presentation attacks

Eye blinking analysis



Defense from face presentation attacks

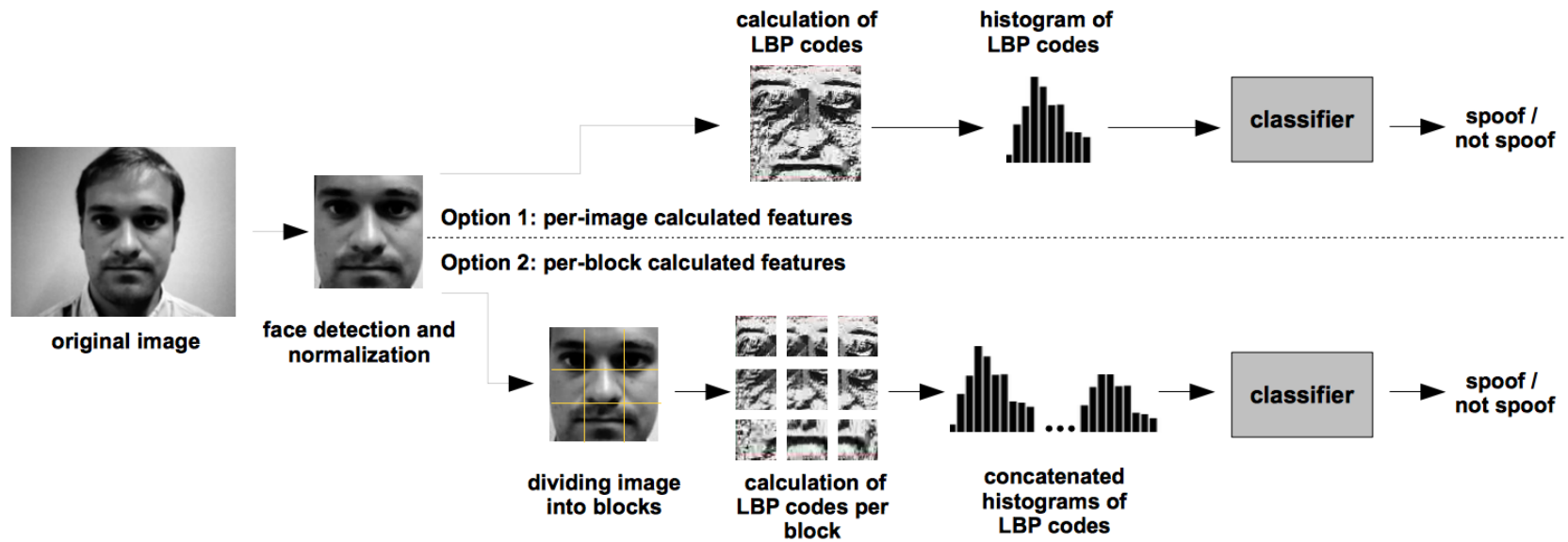
Motion analysis



LBP and optical-flow are used to detect print-attacks and replay-attacks

Defense from face presentation attacks

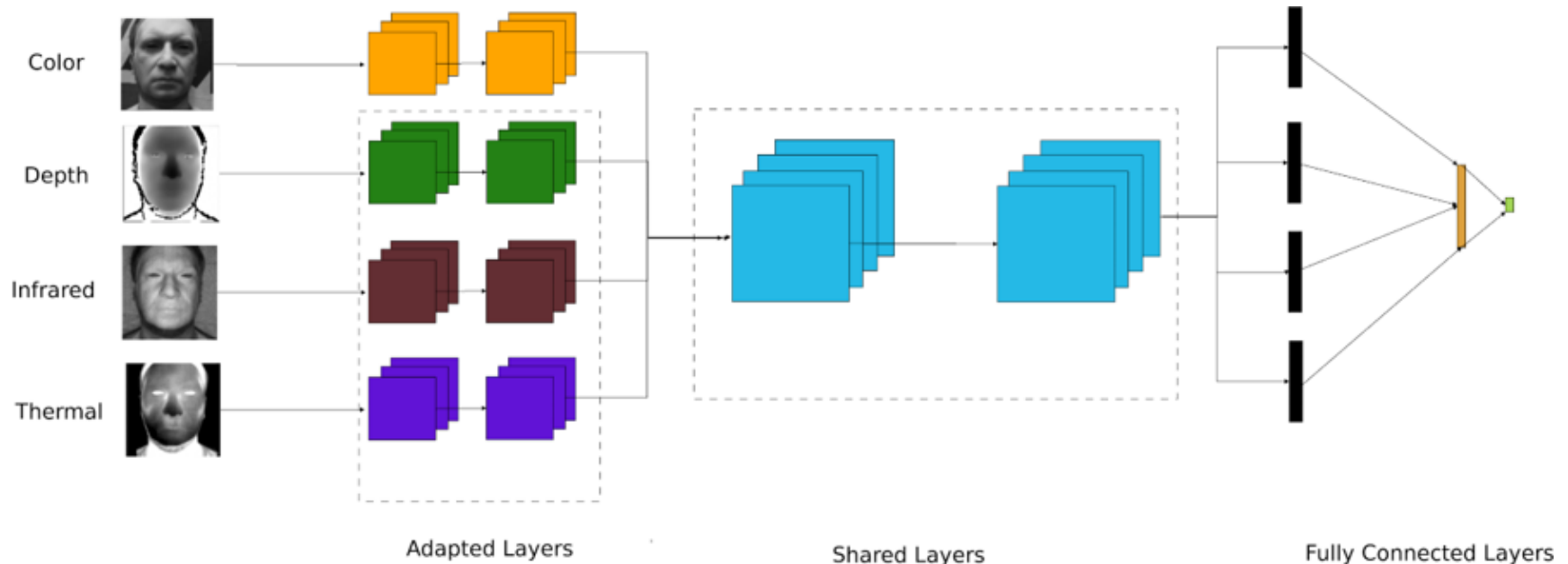
Texture analysis



LBP in 3 dimensions (LBP-TOP) are used to differentiate between real and spoof face presentations

Defense from face presentation attacks

Multi-spectral face presentation attack defense



A multi-channel (VIS, NIR, Depth, TH) CNN-based approach